

Malware Analysis: Debugging and Patching

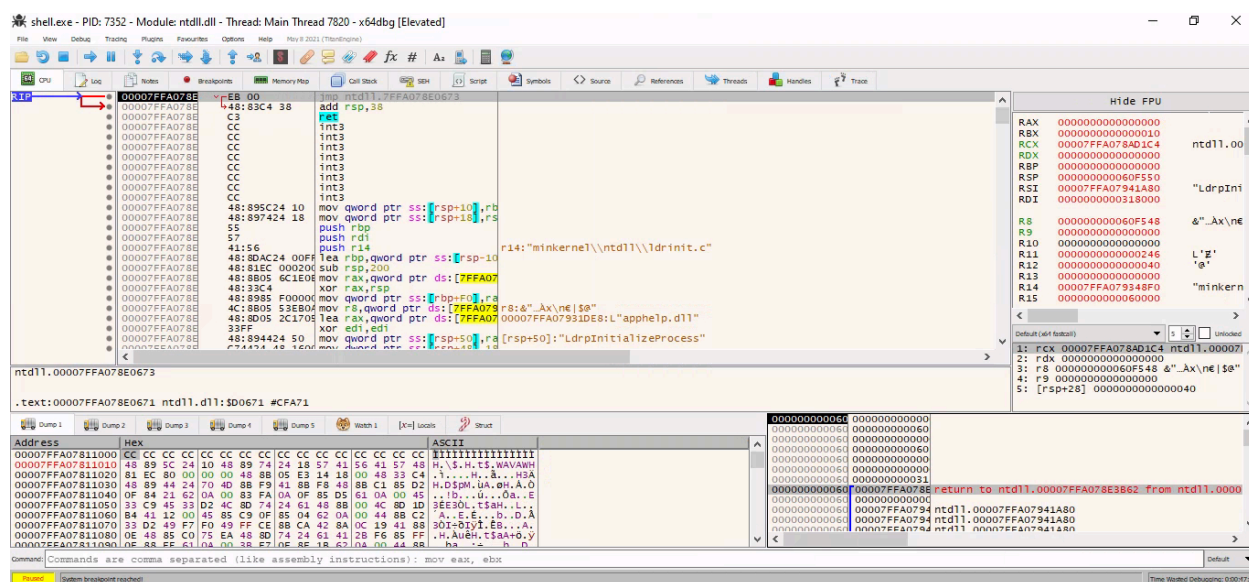
Objectives

The objective of this lab was to use a debugger to modify a piece of malware, preventing it from checking if it is being run in a virtual machine, thus allowing it to automatically proceed with connecting to its command and control (C2) server.

Steps

1. Opened malware in the debugger

The malicious executable was loaded into a debugger to analyze its code and behavior.



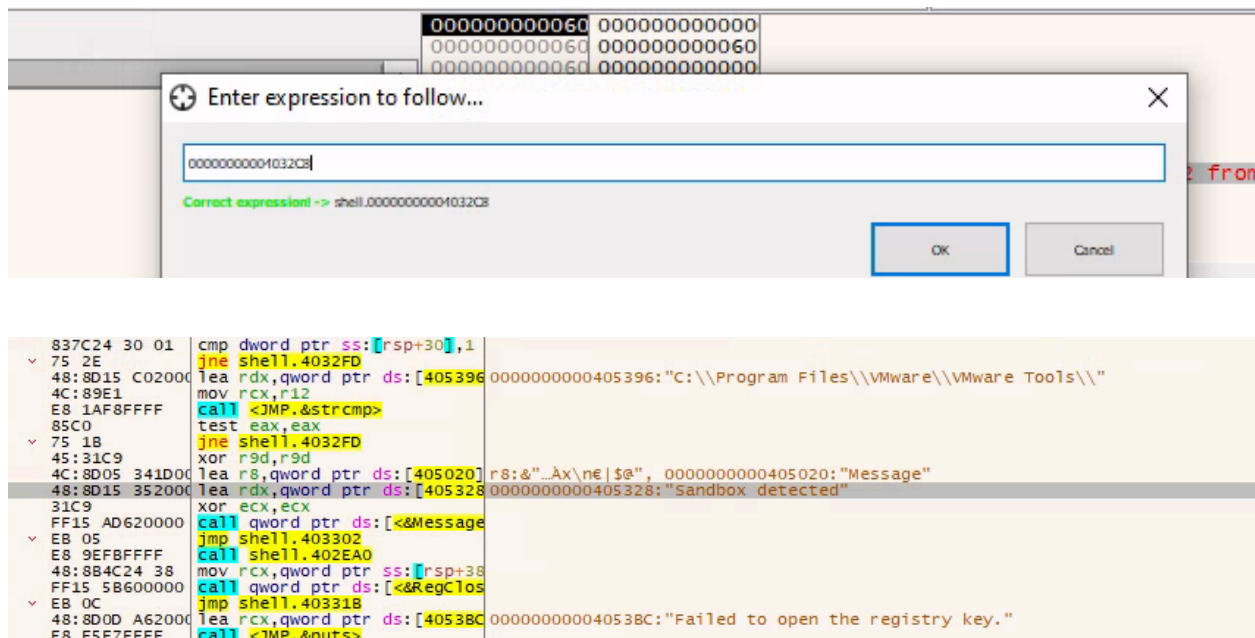
2. Set up INetSim

INetSim was configured to simulate the internet services that the malware would attempt to connect to.

```
[root@htb-u5yvllrkrj]# ./inetsim
INetSim 1.3.2 (2020-05-19) by Matthias Eckert & Thomas Hungenberg
Main logfile '/var/log/inetsim/main.log' does not exist. Trying to create it...
Main logfile '/var/log/inetsim/main.log' successfully created.
Sub logfile '/var/log/inetsim/service.log' does not exist. Trying to create it...
Sub logfile '/var/log/inetsim/service.log' successfully created.
Debug logfile '/var/log/inetsim/debug.log' does not exist. Trying to create it...
Debug logfile '/var/log/inetsim/debug.log' successfully created.
Using log directory: /var/log/inetsim/
Using data directory: /var/lib/inetsim/
Using report directory: /var/log/inetsim/report/
Using configuration file: /etc/inetsim/inetsim.conf
Parsing configuration file.
Configuration file parsed successfully.
=== INetSim main process started (PID 4472) ===
Session ID: 4472
Listening on: 10.10.14.223
Real Date/Time: 2024-06-16 13:27:14
Fake Date/Time: 2024-06-16 13:27:14 (Delta: 0 seconds)
Forking services...
* dns_53_tcp_udp - started (PID 4476)
* time_37_udp - started (PID 4492)
* daytime_13_tcp - started (PID 4493)
```

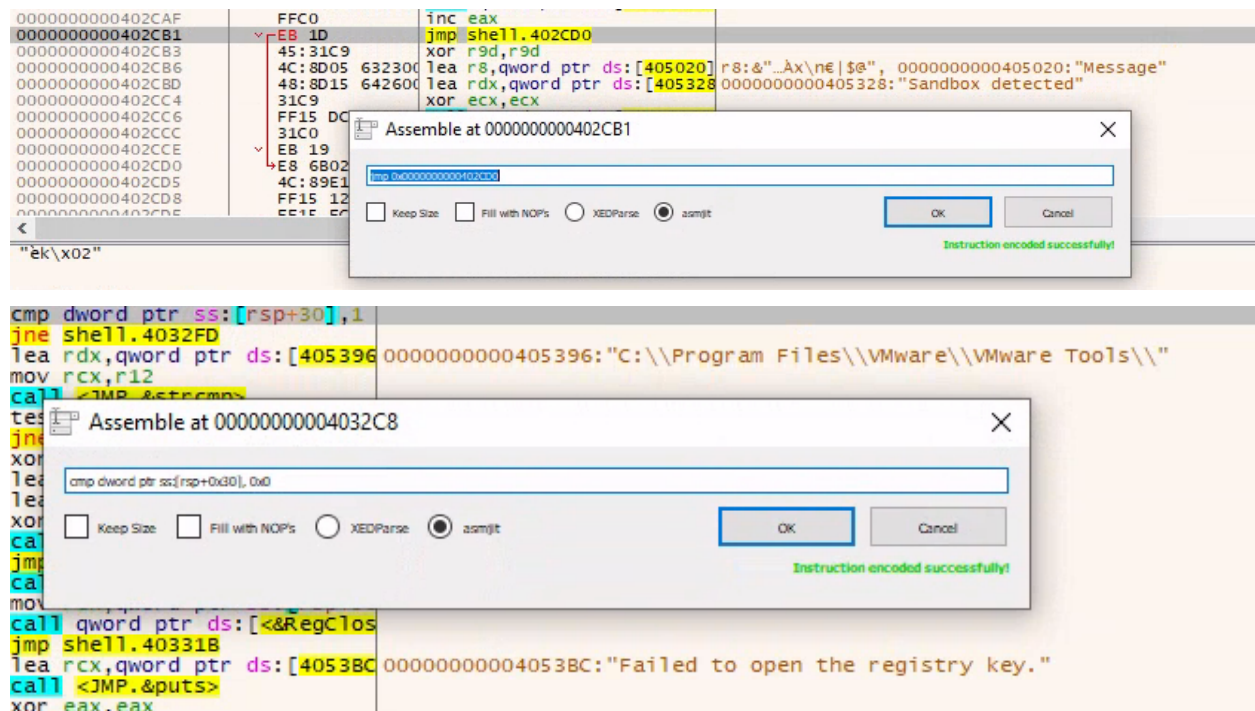
3. Found the address of the code that needs to be patched

Through code analysis, the specific address in the malware's code responsible for virtual machine detection was identified.



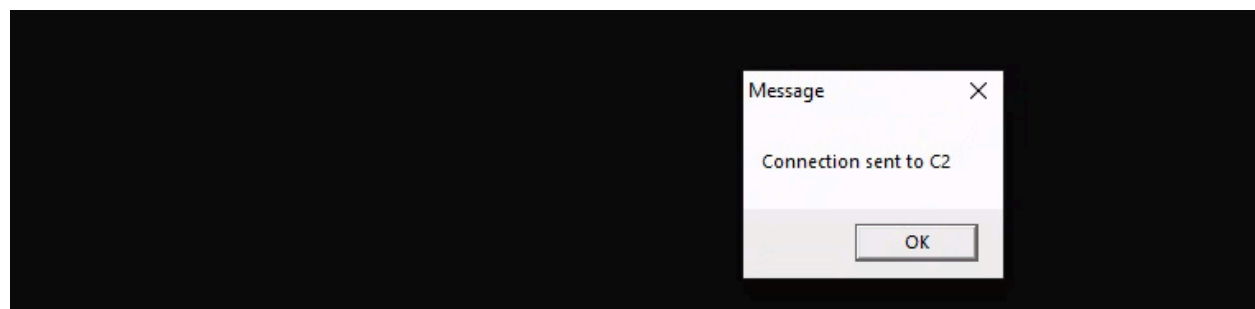
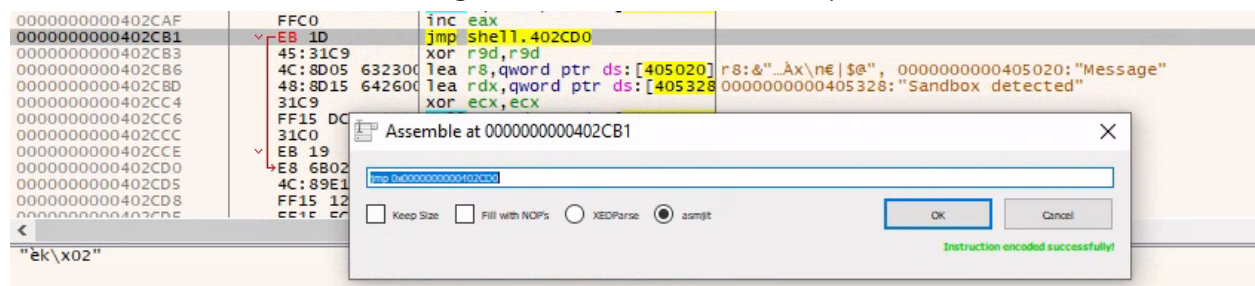
4. Patched the code by changing the values

The identified code was modified to bypass the virtual machine check, enabling the malware to proceed with its operations.



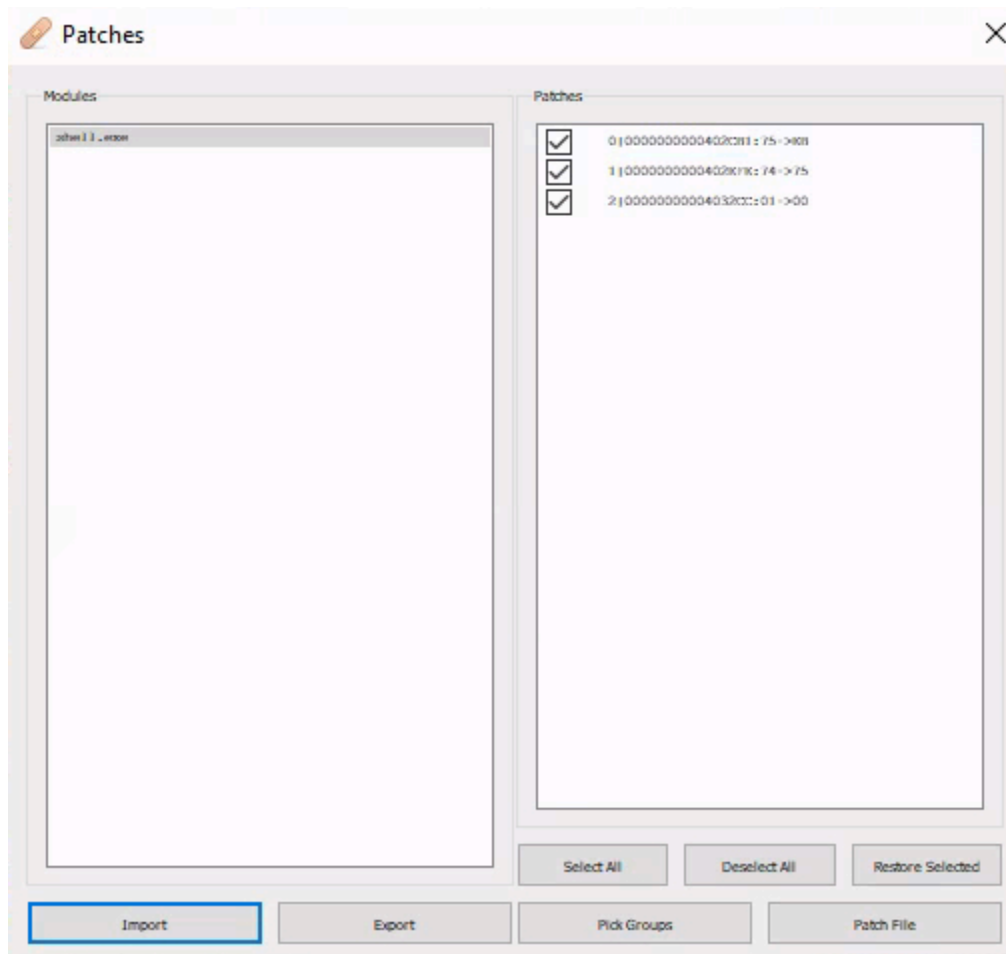
5. Connection was successfully established with the C2

After patching, the malware was executed, and it successfully connected to the simulated C2 server, confirming the effectiveness of the patch.



6. Saved the patch

The changes made to the malware code were saved to ensure the patch was permanent.



7. Analyzed DNS and HTTP traffic that was created by the malware

The network traffic generated by the patched malware was captured and analyzed, focusing on DNS and HTTP requests to understand its communication patterns with the C2 server.

DNS	67	Standard query	0xb3de	AAAA	ms-windows-update.com
DNS	67	Standard query response	0xb3de	AAAA	ms-windows-update.com
DNS	77	Standard query	0x8392	AAAA	settings-win.data.microsoft.com
DNS	77	Standard query response	0x8392	AAAA	settings-win.data.microsoft.com
DNS	69	Standard query	0x7b45	AAAA	ctldl.windowsupdate.com
DNS	69	Standard query response	0x7b45	AAAA	ctldl.windowsupdate.com

```
GET /msdownload/update/v3/static/trustedr/en/authrootstl.cab?c6e01d037218321a HTTP/1.1
Connection: Keep-Alive
Accept: */*
If-Modified-Since: Tue, 25 Apr 2023 17:54:30 GMT
If-None-Match: "46eef7fb9e77d91:0"
User-Agent: Microsoft-CryptoAPI/10.0
Host: ctldl.windowsupdate.com
```

```
HTTP/1.1 405 Method Not Allowed
Server: WebSockify Python/2.7.18
Date: Sun, 16 Jun 2024 13:23:29 GMT
Connection: close
Content-Type: text/html
```

```
<head>
<title>Error response</title>
</head>
<body>
<h1>Error response</h1>
<p>Error code 405.
<p>Message: Method Not Allowed.
<p>Error code explanation: 405 = Specified method is invalid for this resource..
</body>
```

Conclusion

This lab demonstrated the process of using a debugger to patch malware, effectively bypassing its virtual machine checks and allowing it to connect to a C2 server. By setting up a simulated environment with INetSim, the behavior of the patched malware could be safely observed and analyzed. This approach is crucial for understanding malware communication protocols and preparing defenses against such threats.